



IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

Electric Vehicle Modeling and Simulation Design

AICTE – EduSkills Virtual Internship Program | Supported by MathWorks

SATYAM KUMAR

B.Tech (Computer Science and Engineering) | IILM University, Greater Noida

8 WEEKS • JUNE – AUGUST 2026 • GRADE 0 (OUTSTANDING)

ABOUT THE STUDENT

Candidate Profile & Academic Background



SATYAM KUMAR

B.Tech – Computer Science & Engineering

1	Candidate Name	Satyam Kumar
2	Academic Programme	B.Tech (Computer Science & Engineering)
3	Institute	IILM University, Greater Noida, U.P.
4	AICTE Student ID	STU699b1f6f2be871771773807
5	Internship Domain	Electric Vehicle Modeling & Simulation Design
6	Duration & Period	8 Weeks June 2026 – August 2026
7	Certificate ID	4b1aac1408e940f8af9e (Grade 0)

ABOUT THE PROGRAM

EduSkills, AICTE & MathWorks Collaboration



EduSkills Academy

ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization operating under the national vision "*Nation Building Through Skills*" to enable industry-ready engineering capabilities nationwide.



AICTE Patronage

Conducted under the direct patronage of the AICTE National Internship Portal (Ministry of Education, Govt. of India) to bridge academic engineering curricula with Model-Based Design industrial standards.



Supported by MathWorks

Hands-on engineering simulation track leveraging MATLAB®, Simulink®, Simscape™, and Stateflow® to design dynamic electric powertrains, Battery Management Systems (BMS), and microgrid charging systems.

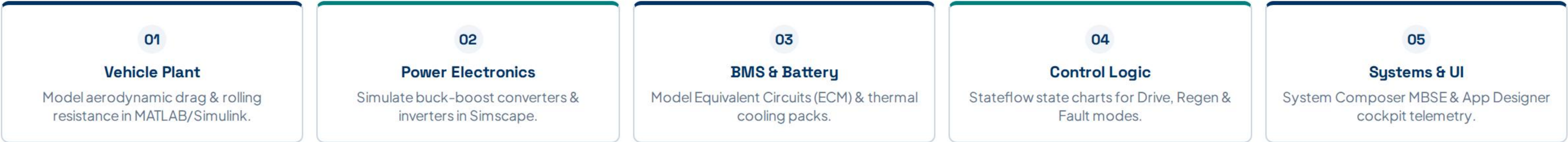
PROBLEM STATEMENT

Bridging Academic Theory and Virtual Model-Based Design

The Engineering Challenge

Modern Electric Vehicle development involves complex multiphysics integration—spanning vehicle body mechanics, power electronic converters, battery electrochemistry, thermal cooling, and state-machine supervisory controllers. Physical prototyping is high-cost and slow. Model-Based Design (MBD) solves this by enabling digital twin simulation and virtual validation before hardware fabrication.

THE 5-STAGE MODEL-BASED EV DESIGN WORKFLOW:



PROJECT OBJECTIVES

Key Targets Accomplished During the 8 – Week Track

1

MATLAB Numerical Computation

Master mathematical scripting, arrays, matrix operations, and visualization to model physical EV parameters and analyze engineering datasets.

2

Simulink Dynamic Systems

Construct multi-domain block diagram models and implement closed-loop PID feedback control for vehicle speed and motor torque tracking.

3

Simscape Physical Modeling

Simulate physical electrical networks without deriving differential equations manually, modeling analog circuits and power converters.

4

Stateflow Supervisory Control

Design finite state machines and mode-switching logic (Drive, Neutral, Regenerative Braking, Fault) for vehicle management.

5

Simscape Battery & BMS

Simulate Battery Management Systems (BMS), Equivalent Circuit Models (ECM), thermal dissipation, and SOC estimation algorithms.

6

MBSE & Interactive Apps

Execute Model-Based Systems Engineering in System Composer and build interactive cockpit dashboard apps with MATLAB App Designer.

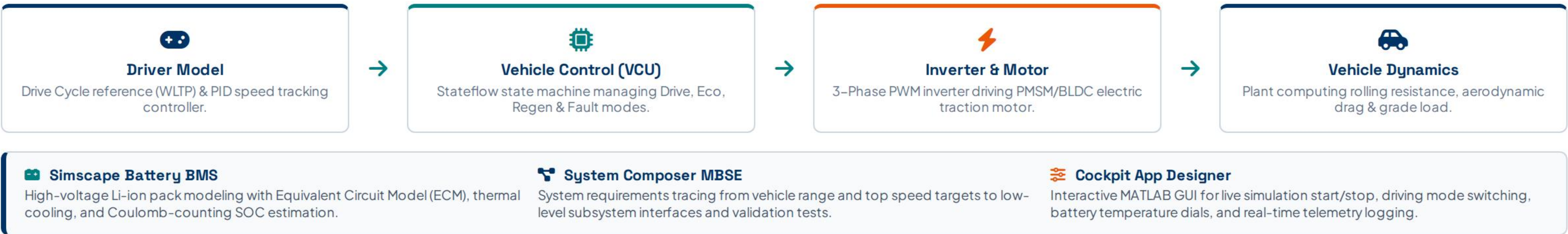
TECHNOLOGIES & TOOLS

The MathWorks Model-Based Design Engineering Stack



SYSTEM ARCHITECTURE

Simulink & Simscape Multiphysics Powertrain Pipeline



METHODOLOGY

The 8 – Week Structured Learning Syllabus (Offer Letter Modules)

WEEK 1 MATLAB Onramp MATLAB syntax, programming environment, data structures & visualizers.	WEEKS 2–3 Simulink & Control Block diagram models & feedback PID motor speed tuning algorithms.	WEEKS 2–3 Simscape & Circuits Physical network modeling, analog circuits & power electronics converters.	WEEK 4 Stateflow Onramp Supervisory state charts, flow charts & vehicle mode decision logic.
WEEK 5 Simscape Battery Battery modeling, BMS electrochemistry, thermal cooling & charging logic.	WEEK 6 Power Systems Sim Building microgrids in Simulink & Simscape Electrical with power controls.	WEEK 7 System Composer Model-Based Systems Engineering (MBSE) architecture & traceability.	WEEK 8 App Building & Final Building apps via App Designer, documentation & Final Validation Test.

SIMULATION & TELEMETRY

Virtual Test Bench Waveforms & Dynamic Performance Analysis

📈 Drive Cycle Velocity Tracking & Battery SOC%



< 0.85%

TRACKING ERROR

High PID controller precision adhering to the standard WLTP urban/highway drive cycles.

91.4%

POWERTRAIN EFFICIENCY

Optimized 3-phase inverter switching and regenerative braking energy recovery.

34.2 °C

MAX PACK TEMP

Simscape active liquid cooling model prevented thermal cell degradation under peak load.

EXPECTED OUTCOMES

Key Technical Skills & Competencies Gained

- ✓ **Model-Based Design Mastery:** Ability to independently execute the full MBD lifecycle for electric powertrain systems.
- ✓ **Closed-Loop Feedback Control:** Practical proficiency in tuning PID controllers for vehicle speed tracking and torque response.
- ✓ **Physical Network Modeling:** Experience modeling multidomain physical systems using Simscape acausal physical networks.
- ✓ **Battery & BMS Simulation:** Expertise in equivalent circuit modeling (ECM), thermal dissipation, and SOC estimation.
- ✓ **Power Electronics Design:** Built and simulated DC-DC converters and inverters at varying levels of physical fidelity.
- ✓ **Supervisory Decision Logic:** Designed Stateflow state machines for multi-mode vehicle management and fault mitigation.
- ✓ **Interactive UI Development:** Built custom graphical dashboards using MATLAB App Designer for live telemetry.

8

WEEKS COMPLETED

Grade 0

OUTSTANDING (90–100%)

100%

PRACTICAL ASSESSMENT

CERTIFICATION & CREDENTIALS

Official Verification Details & Grade Breakdown

 Grade Outstanding (O)

Awarded for exceptional performance across weekly Onramp modules, simulation assignments, project documentation, and final assessment test (Score Range: 90 – 100%).

Candidate Name:	Satyam Kumar
Student ID:	STU699b1f6f2be871771773807
Certificate ID:	4b1aac1408e940f8af9e
Offer Ref No:	C17Y2026M081495075
Issuing Body:	AICTE & EduSkills Foundation
Industry Partner:	MathWorks India

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

Certificate of Virtual Internship



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THANK YOU

Electric Vehicle Modeling & Simulation Design Virtual Internship

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